Fiscal Year:	FY 2005	**Task Last Updated:**	FY 10/24/2005
PI Name:	Doerr, Harold K. M.D.		
Project Title:	Medical Operational Support Team (MOST)		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Smart Medical Systems Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77030	**Congressional District:**	25
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-01
Start Date:	10/01/2002	**End Date:**	10/31/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	2
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Thomson, William (Baylor College of Medicine") Hebl, Michelle (Rice University)		
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
	The specific aims for the Medical Operational Support Team (MOST) project were first articulated in the initial application for funding (submitted in 2001). A multidisciplinary team, known as the Medial Operational Support Team (MOST) was created and is currently being maintained as the key players in the project. The team adapted a Human Patient Simulator (HPS) for use in physiologic scripting and modeling states specific to the aerospace environment. This model was used to help develop, evaluate, and validate medical procedures for use in space. To date, the ground component has been completed for several procedures. Validation testing during first parabolic flights and finally in space remains to be done. The MOST project has supported educational outreach activities since its inception. The MOST team is currently developing an integrated, multimedia medical curriculum. Following validation, it will be initiated as a pilot program for students at NASA-Johnson Space Center. We are in the process of writing up the results of the MOST activities for publication in peer-reviewed journals. The developmental nature of earlier activities delayed		

Task Description:	publication of much of the work. Similarly, transfer of technologies to the private sector cannot commence until validation of the work's effectiveness. These specific aims nicely align with items identified in NASA's Critical Path Roadmap (CPR), including primary risk area numbers 20 (Major illness and Trauma) and 25 (Skill Determination and Training). The need for strategies to address medical emergencies in space environments was reemphasized in a recent NASA report on Guidelines and Capabilities for Designing Human Missions. With regard to anticipated medical events in space, the report stated that "…needs for medical equipment, crew training, and ground support are considerable." The work completed thus far through the MOST project demonstrates significant progress in creating simulation-based training system and physiologic models that are appropriate for medical emergencies that occur in space. Finally, the MOST project has tested, through parabolic flight-based experiments, systems for deploying and using medical emergency equipment in microgravity situations. More work in this area is needed in order to establish an environment in which medial emergencies can be managed effectively through a combination of on-board human and technology resources and ground-based guidance and support. This year's efforts have been positive, with clear and useful products deriving from the work. The next phase of effort for the MOST project involves final testing of interventions for on–board emergencies and completion of the first curriculum for training personnel in these interventions. Our goal is to assist in the creation of the safest possible environment for space travelers who are at risk of acute health problems and/or injury. We focus on those issues identified through NASA reports and related publications and ensure that our efforts incorporate appropriate learning theory. In this way, we can enhance their effectiveness in promoting acquisition of the essential knowledge and skills for timely responses to medical emergencies in relevant environments. As noted by Knowles, considered by many to be the father of adult learning theory, instruction of adults must involve learner-centered methods, as compared to the teacher-centered methods typically used in instruction of children. The challenge for the next phase of the MOST project is to convert the advances made in instructional systems development and testing to an instructional curriculum appropriate to the needs of our audience – whether in a terrestrial or extraterrestrial environment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research in astronaut and flight surgeon training has obvious applications to terrestrial medical training. The same philosophy of operationally-relevant simulation-based training can be applied to the education programs of both traditional and non-traditional health care providers. In particular, our research into sustainment training will establish the standard for this type of testing. In addition, techniques developed by this team can be adapted by terrestrial health care providers; for example, adaptation of the free-floating intubation technique has been evaluated by paramedics as an alternative method for difficult airways. Modeling results are currently being used to create additional simulator-based patients for use in training anesthesiology students.
Task Progress:	The work completed to date in the MOST project reflects significant progress toward development of an integrated, simulation-based training program to prepare flight and ground-based personnel to cope effectively with several anticipated health-related emergencies. Reviewed physiological research data, scripting and modeling literature Scripted HPS physiological models for astronaut subpopulations. Developed, evaluated and validated, in a combination of ground-based analog test environments and during parabolic flight, several procedures relevant to space medicine using the enhanced, space adapted HPS Demonstrated the utility of flight-ready HPS Gathered preliminary intubating laryngeal mask airway (ILMA) insertion data in both restrained and free-floating patients Tested methods for unstowing the on-board defibrillator Evaluated non-clinicians' ability to conduct an in-flightpapillary assessment Evaluated the setup of intravenous fluid infusion system Implemented and assessed the efficacy of the space-adapted HPS system as a training aid for knowledge and skill acquisition of procedures Validated skill set training for surgical and non-surgical procedures (e.g., airway management as well as pneumothorax) using the HPS model Developed training regimens for specific medical procedures (e.g., administering anesthesia to patients in the field) in response to requests from NASA personnel Provided refresher training in emergency medial procedures to NASA personnel.
Bibliography Type:	Description: (Last Updated: 10/22/2008)
Presentation	Bacal, K., Doerr, H. K. "Multi-Discipline Integration of Simulation Technology (Presentation at IMMS 04)" N/A Jan-2004
Presentation	Bacal, K., Doerr, H. K., Miller, R. M. "Communicating Medical Needs to Non-medical Managers (#1 of 4 for Panel #1 at AsMA 04)" N/A May-2004
Presentation	Bacal, K., Hurst IV, V. W., Doerr, H. K. "Medical Scenarios Relevant to Space Flight (#4 of 4 for Panel #1 at AsMA 04)" N/A May-2004
Presentation	Bacal, K., Miller, R., Doerr, H. K. "Scripting Scenarios for the Human Patient Simulator (#2 of 4 for Panel #1 at AsMA 04)" N/A May-2004
Presentation	Doerr, H. K. "Future of Simulation: Virtual Reality, New Models, Other Disciplines (Panel at IMMS 04)" N/A Jan-2004
Presentation	Doerr, H. K., Bacal, K, Hurst IV, V "Teamwork in Space Medicine (Panel at NYLF 04)" N/A Jul-2004
Presentation	Doerr, H. K., Bacal, K. "Mars - It's closer than you think: Space Medicine using Technology, Simulation and Ideas for the Future" N/A Jan-2004

Presentation	Doerr, H. K., Bacal, K., Hurst IV, V. "Creating a Scenario Suitable for Multiple Caregivers (#3 of 4 for Panel #2 at AsMA 04)" N/A May-2004
Presentation	Doerr, H. K., Clark, J. B. "Introduction to the Medical Operational Support Team (Panel #2 at AsMA 04; #1 of 4 for Panel #2)" N/A May-2004
Presentation	Doerr, H. K., Mc Swain, N. E. "Simulation Training for Adminstering Anesthesia to Astronauts following Space Flight (#4 of 4 for Panel #2 at AsMA 04)" N/A May-2004
Presentation	Hurst IV, V. W. "Simulations for the Novel Environment (Panel #1 at AsMA 04)" N/A May-2004
Presentation	Hurst IV, V. W., Bacal, K., Doerr, H. K. "The Changing Personnel of Simulation Centers (Presentation at IMMS 04)" N/A Jan-2004
Presentation	Hurst IV, V. W., Doerr, H. K., Bacal, K. "Creating Simulated Microgravity Patient Models (#2 of 4 for Panel #2)" N/A May-2004
Presentation	Miller, R., Bacal, K., Doerr, H. K. "Developing a Simulation-based Training Program for Non-traditional Caregivers (#3 of 4 for Panel #1 at AsMA 04)" N/A May-2004